Docker Images

How to create your own Image?

Manual approach:

1. OS - Ubuntu
2. Update apt repo
3. Install dependencies using apt
4. Install Python dependencies using pip
5. Copy source code to /opt folder
6. Run the web server using “flask” command

Quick overview:

Dockerfile

```
FROM Ubuntu

RUN apt-get update
RUN apt-get install python

RUN pip install flask
RUN pip install flask-mysql

COPY . /opt/source-code

ENTRYPOINT FLASK_APP=/opt/source-code/app.py flask run
```

To create locally on the system:

```
docker build Dockerfile -t mmumshad/my-custom-app
```

To make it available in public registry:

```
docker push mmumshad/my-custom-app
```

Docker file is the text file, written in format that docker could understand
Syntax: Instruction Argument

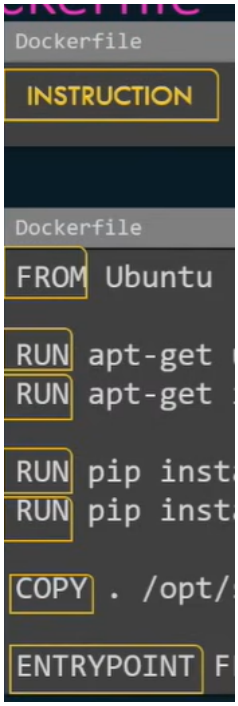
Dockerfile

INSTRUCTION ARGUMENT

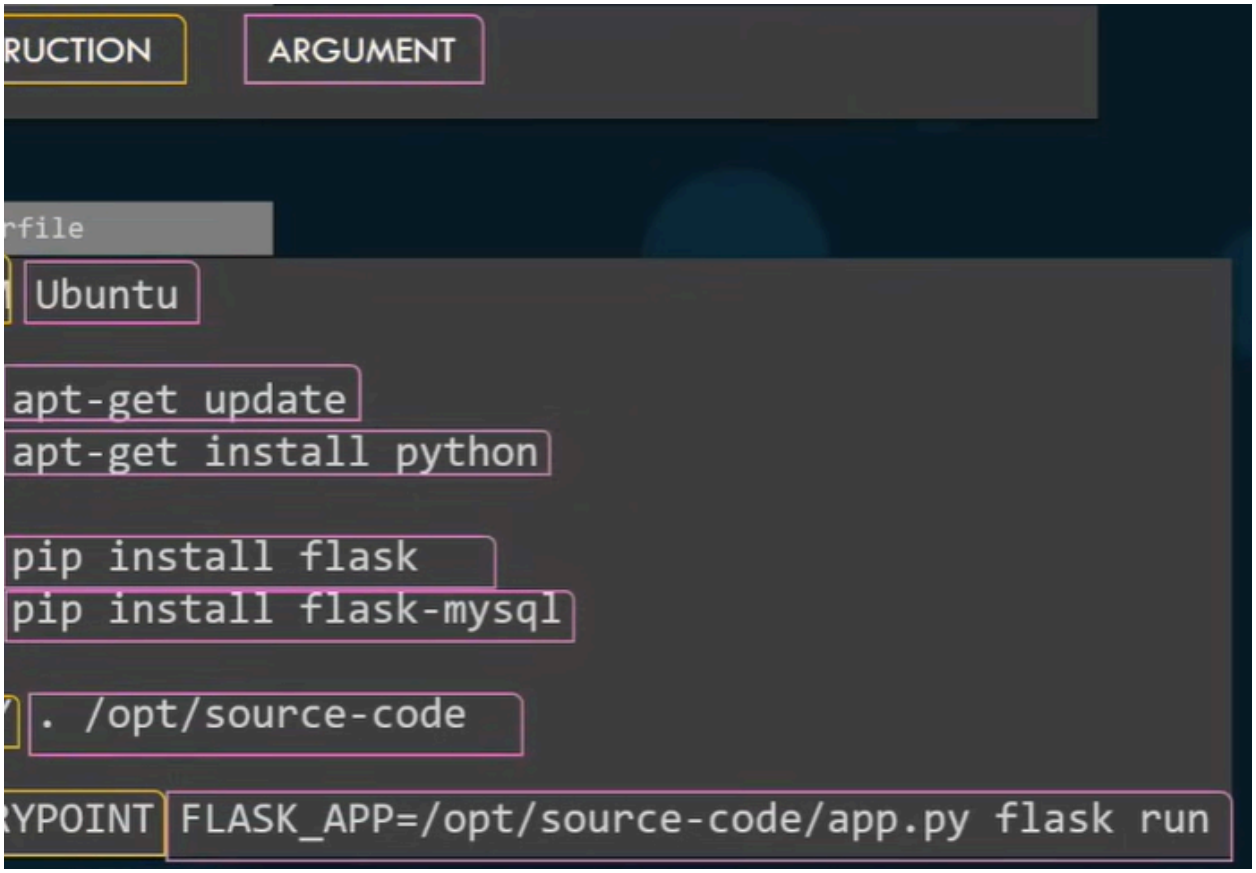
Dockerfile

```
FROM Ubuntu

RUN apt-get update
RUN apt-get install python

RUN pip install flask
RUN pip install flask-mysql

COPY . /opt/source-code

ENTRYPOINT FLASK_APP=/opt/source-code/app.py flask run
```

Instructions:



Arguments:



Start writing from Base OS or another image:

```
FROM Ubuntu
```

Start from a base OS or another image

Run command to install it:

```
RUN apt-get update
RUN apt-get install python

RUN pip install flask
RUN pip install flask-mysql
```

Install all dependencies

Copy command for copying file from local system to the docker image, current folder(.) to path(/opt/source-code)
Entrypoint, image is run as a container

```
COPY . /opt/source-code

ENTRYPOINT FLASK_APP=/opt/source-code/app.py flask run
```

Copy source code

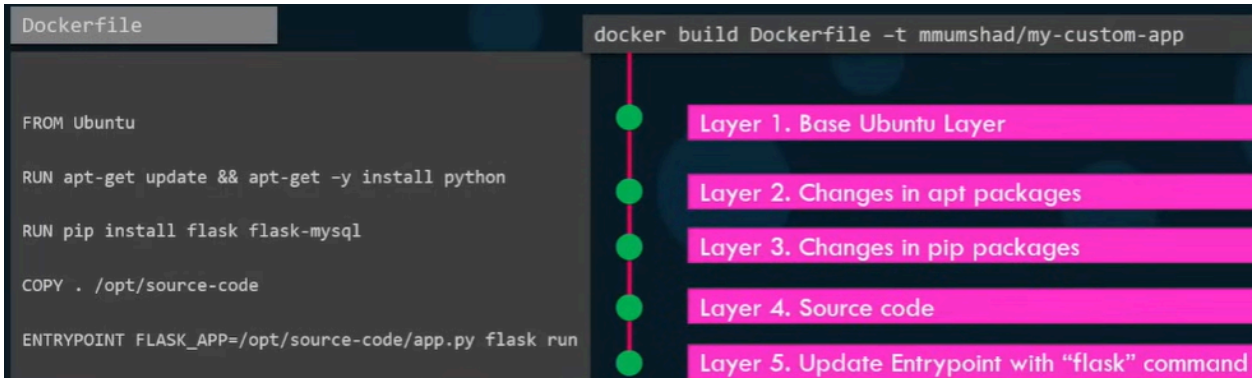
Specify Entrypoint

Layered Architecture:

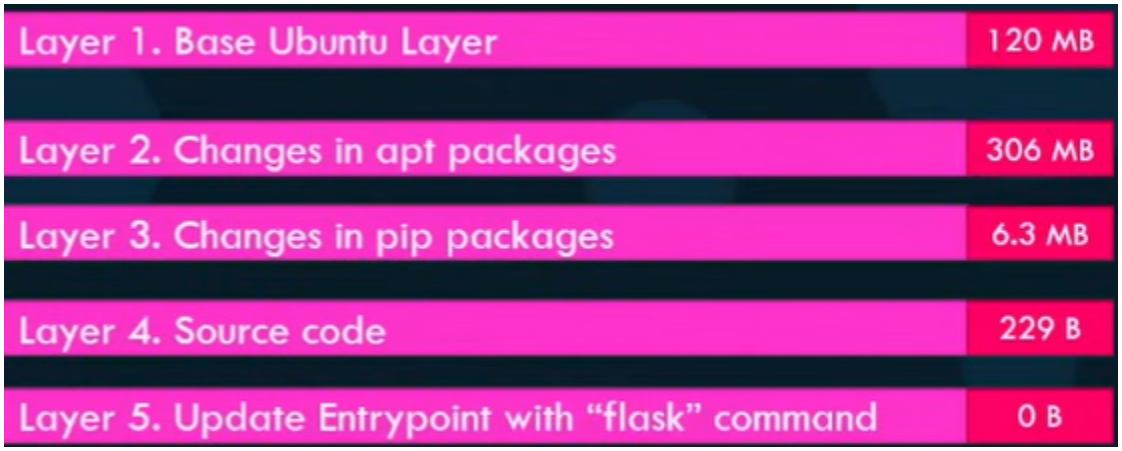
Dockerfile

```
FROM Ubuntu

RUN apt-get update && apt-get -y install python

RUN pip install flask flask-mysql

COPY . /opt/source-code

ENTRYPOINT FLASK_APP=/opt/source-code/app.py flask run
```

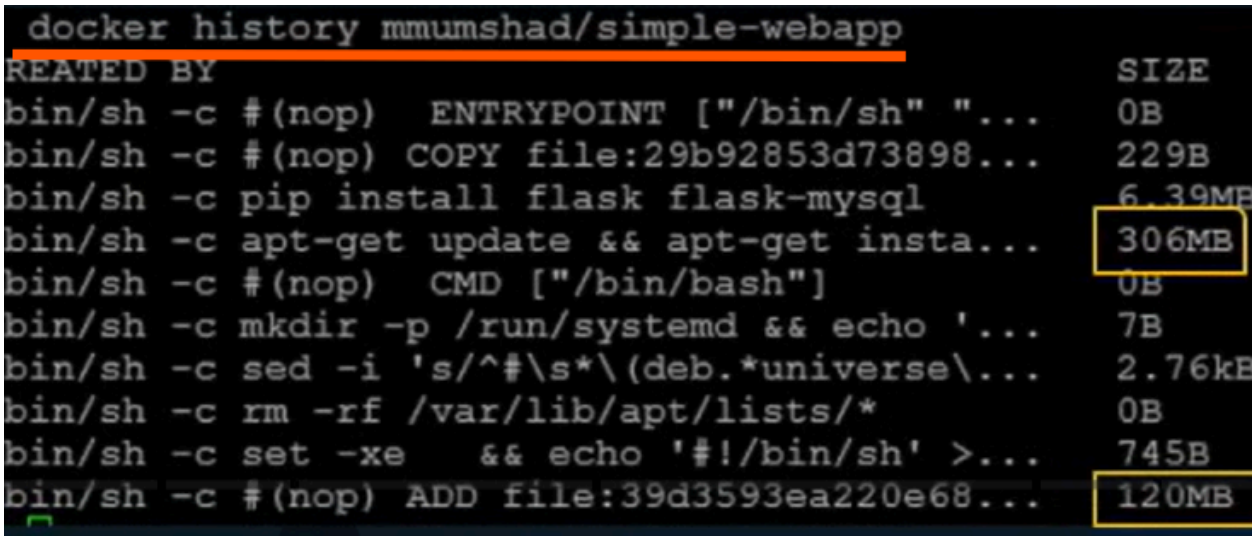
```
docker build Dockerfile -t mmumshad/my-custom-app
```



Each layer stores the changes from the previous layer, it is reflected in the side as well.



Use the docker history command to check the size.

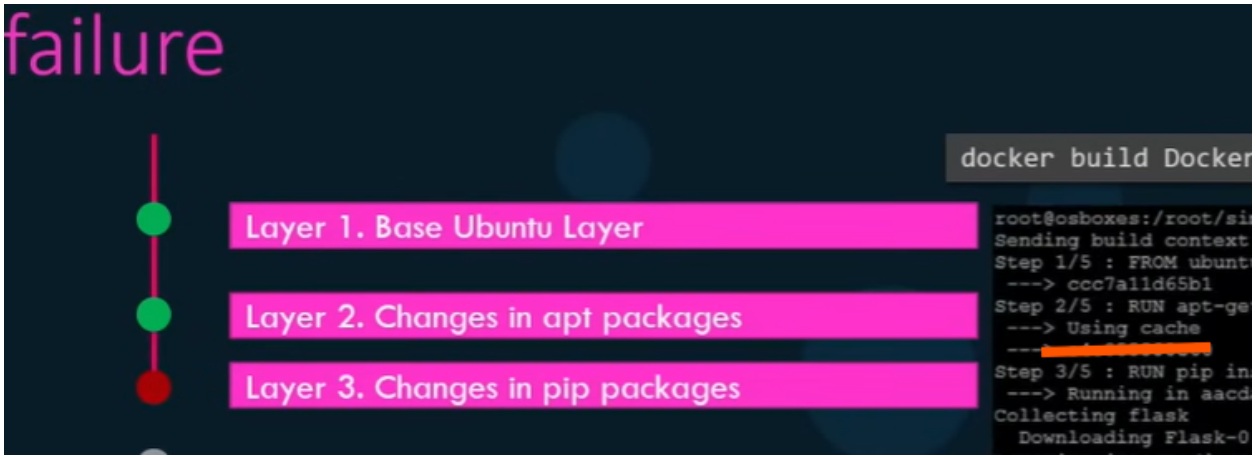


Output after running docker build command,
Command: **docker build .**

```
root@osboxes:/root/simple-webapp-docker # docker build .
Sending build context to Docker daemon  3.072kB
Step 1/5 : FROM ubuntu
 ---> ccc7a11d65b1
Step 2/5 : RUN apt-get update && apt-get install -y python python-setuptools python-dev
 ---> Running in a7840dbfad17
Get:1 http://archive.ubuntu.com/ubuntu xenial InRelease [247 kB]
Get:2 http://security.ubuntu.com/ubuntu xenial-security InRelease [102 kB]
Get:3 http://archive.ubuntu.com/ubuntu xenial-updates InRelease [102 kB]
Get:4 http://security.ubuntu.com/ubuntu xenial-security/universe Sources [46.3 kB]
Get:5 http://archive.ubuntu.com/ubuntu xenial-backports InRelease [102 kB]
Get:6 http://security.ubuntu.com/ubuntu xenial-security/main amd64 Packages [440 kB]
Step 3/5 : RUN pip install flask flask-mysql
 ---> Running in a4a6c9190ba3
Collecting flask
  Downloading Flask-0.12.2-py2.py3-none-any.whl (83kB)
Collecting flask-mysql
  Downloading Flask_MySQL-1.4.0-py2.py3-none-any.whl
Removing intermediate container a4a6c9190ba3
Step 4/5 : COPY app.py /opt/
 ---> e7cdab17e782
Removing intermediate container faaaaf63c512
Step 5/5 : ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0
 ---> Running in d452c574a8bb
 ---> 9f27c36920bc
Removing intermediate container d452c574a8bb
Successfully built 9f27c36920bc
```

```
docker build Dockerfile -t mmumshad/my-custom-app
```

If the third layer is failed, then it will reuse the previous layer from cache



And continue to build remaining layers.

failure

- Layer 1. Base Ubuntu Layer
- Layer 2. Changes in apt packages
- Layer 3. Changes in pip packages
- Layer 4. Source code
- Layer 5. Update Entrypoint with “flask” command

```
docker build Docker

root@osboxes:/root/si
Sending build context
Step 1/5 : FROM ubunt
 ---> ccc7a11d65b1
Step 2/5 : RUN apt-ge
 ---> Using cache
 ---> e4c055538e60
Step 3/5 : RUN pip in
 ---> Running in aacd
Collecting flask
  Downloading Flask-0
Removing intermediate
Step 4/5 : COPY app.p
 ---> af41ef57f6f3
Removing intermediate
Step 5/5 : ENTRYPOINT
 ---> Running in 3d74
 ---> 910416d360b6
Removing intermediate
Successfully built 91
```